From Text to Signals in Finance

Lecture 1 — Introduction

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June 20, 2026

Outline

The Founding Moment: Tetlock (2007)

The Setup

Autumn 2006. Paul Tetlock assembles a dataset:

- Daily fraction of negative words in the *WSJ* "Abreast of the Market" column (1984–1999)
- Next-day Dow Jones Industrial Average return

Results:

- High negativity ⇒ lower next-day returns [puzzling under EMH]
- Effect *reverses* over the subsequent week ⇒ temporary mispricing
- High/low negativity ⇒ high next-day trading volume

THE BIGGER PICTURE

Human language can be turned into a quantitative return-predicting signal. The reversal tells us it reflects sentiment, not fundamental news.

Dictionary Methods: Harvard IV-4 and LM

General Inquirer (Stone et al., 1966)

- 11,000 word stems classified into Positive, Negative, Strong, Weak, …
- Score a document: count words per category, normalise by length
- Transparent and reproducible

Loughran–McDonald (LM, 2011)

- *tax*, *liability*, *depreciation* flagged "negative" by Harvard IV-4 but are merely financial jargon
- LM dictionary hand-coded from 10-K filings: Negative, Positive, Uncertainty, Litigious, Strong Modal, Weak Modal
- Substantially outperforms Harvard IV-4 in predicting filing-date abnormal returns

Fundamental limitation of all dictionary methods

Static, order-invariant. "Earnings did *not* decline" = "Earnings *did* decline" unless negation is hard-coded.

Sources of Financial Text

Earnings calls Scripted presentation + unscripted Q&A; Q&A is spontaneous and more information-rich

10-K / 10-Q filings MD&A, risk factors; up to 200 pages; primary source for Loughran–McDonald (2011)

Newswire / news High frequency; narrow exploitation window; value in *speed* and *breadth*

Analyst reports Informationally dense; linguistic complexity $\propto$ information asymmetry

Central bank comms FOMC minutes and statements move rates and equity prices on release

Social media High-frequency, noisy; predictive power documented but robustness debated

Why Text Signals Survive Market Efficiency

Semi-strong EMH (Fama, 1970): prices reflect all publicly available information. Yet large empirical literature finds text predicts returns. Three reconciling mechanisms:

1. **Processing costs & limited attention**
   A 200-page 10-K is costly to read. Not all investors process it fully. NLP at scale makes it feasible ⇒ anomaly should attenuate with technology.
2. **Soft information & interpretive disagreement**
   Cautious tone = genuine uncertainty *or* expectation management? Disagreement ⇒ prices reflect average of beliefs, not the most informative one.
3. **Latent information**
   Regulation FD limits explicit selective disclosure. Information leaks through word choice, omissions, and rhetorical structure.

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Text analysis adds value not because markets are irrational, but because *information extraction from text is genuinely costly*.

Four Empirical Regularities

The theory must explain these stylised facts:

1. **Negative tone $\Rightarrow$ negative short-run returns, with reversal.**
   Tetlock (2007): pessimistic WSJ prose $\rightarrow$ lower next-day DJIA, then mean-reversion.
2. **Financial-domain dictionaries outperform general-domain ones.**
   Loughran–McDonald (2011): LM Negative $\gg$ Harvard IV-4 for predicting filing-date abnormal returns.
3. **Earnings call tone predicts future earnings surprises.**
   Larcker–Zakolyukina (2012): deceptive CEO/CFO speech has distinct linguistic markers.
4. **LLM signals show incremental predictive power.**
   Lopez-Lira & Tang (2023): ChatGPT sentiment for headlines predicts next-day returns; absent for GPT-1 and BERT $\Rightarrow$ emergent property of scale.

The Textual Signal: Formal Definition

Definition

A **textual signal** is a measurable function

$$f : \mathcal{D} \longrightarrow \mathbb{R}^k$$

mapping document d_t (at time t) to signal value $s_t = f(d_t)$.

Examples of f:

- Fraction of LM Negative words ($k = 1$)
- Vector of LDA topic proportions ($k =$ number of topics)
- Word2Vec sentence embedding ($k = 300$)
- LLM classifier output ($k =$ number of classes)

This course = investigating the best choice of f: from simple word counts $\rightarrow$ topic models $\rightarrow$ transformer embeddings $\rightarrow$ LLMs as reasoning agents.

The Linear Factor Model and Economic Significance

The central empirical question:

$$\mathrm{Cov}(s_t, r_{t+1}) \neq 0\ ?$$

Under a linear factor model:

$$r_{t+1} = \alpha + \beta^\top s_t + \varepsilon_{t+1}, \qquad \mathbb{E}[\varepsilon_{t+1} \mid s_t] = 0$$

β is **economically significant** if and only if:

1. $|\beta| > 0$ after controlling for size, value, momentum, ...
2. The implied Sharpe ratio exceeds plausible transaction costs

Both conditions are necessary

Statistical significance without economic significance does not justify a trading strategy.

Identification Challenges

Showing $\text{Cov}(s_t, r_{t+1}) \neq 0$ is necessary but *not sufficient*:

Reverse causality Document d_t may be written *in response to* lagged returns, not in anticipation of future ones.

Confounding Macro conditions drive both tone and returns. Use high-frequency event studies with known release times.

Multiple testing NLP produces thousands of candidate signals. Use strict temporal out-of-sample split (train pre-2010, test post-2010).

Model misspecification The choice of f is itself an assumption. Dictionary f assumes only word counts matter; LLM f relaxes this but introduces training-data biases.

Tokenisation and the Preprocessing Pipeline

Before computing any representation, raw text is standardised:

1. **Case folding** — "Revenue" $\equiv$ "revenue"
2. **Stop-word removal** — remove "the", "and", "of", … (100–300 words)
3. **Stemming / lemmatisation**
   - Stemming: *earnings, earned, earns* $\rightarrow$ *earn*
   - Lemmatisation: *better* $\rightarrow$ *good* (more accurate, slower)
4. **Vocabulary truncation** — retain top V' types; discard very rare words

Result: finite vocabulary $\mathcal{V}$ of size V (typically 10 000–100 000)

Note: LLMs use *sub-word tokenisation* — "earnings" $\rightarrow$ ["earn", "ings"] — allowing rare words to be represented as combinations of frequent sub-units.

Bag-of-Words Model

Definition

$$\mathbf{x}(d) = \big(\text{count}(v_1, d), \ldots, \text{count}(v_V, d)\big)^\top \in \mathbb{Z}_{\geq 0}^{V}$$

Word order is discarded. The document is treated as an unordered multiset.

Virtues

- Simple and interpretable
- Sparse — efficient storage (CSR/CSC)
- Competitive baseline for many tasks

The fundamental deficiency

- "The fund outperformed the market" = "The market outperformed the fund" (identical BoW vectors, opposite meanings)

The term-document matrix $\mathbf{X} \in \mathbb{Z}_{\geq 0}^{N \times V}$ stacks BoW vectors for all N documents.

One-Hot Encoding: The Discrete Metric Problem

Each token v_k maps to the standard basis vector:

$$\mathbf{e}_k = (\underbrace{0,\ldots,0}_{k-1}, 1, \underbrace{0,\ldots,0}_{V-k})^\top \in \{0,1\}^V$$

BoW = sum of one-hot vectors of constituent tokens.

UNDER THE HOOD

The problem: orthogonality.

$$\mathbf{e}_j^\top \mathbf{e}_k = \mathbb{1}[j=k] \quad \Rightarrow \quad \mathrm{sim}(\text{revenue, income}) = 0 = \mathrm{sim}(\text{revenue, penguin})$$

Every pair of distinct words is equally dissimilar — the *discrete metric*.
This is semantically incorrect: financial analysts know that "revenue" and "income" are nearly synonymous in most contexts.

Word embeddings replace the discrete metric with a continuous geometry that reflects distributional similarity.

TF-IDF: Motivation and Definitions

Raw BoW counts dominated by stop words. TF-IDF weights by document-level rarity:

$$\underbrace{\mathrm{tf}(v,d)}_{\text{term freq.}} = \frac{\mathrm{count}(v,d)}{|d|} \qquad \underbrace{\mathrm{idf}(v,\mathcal{C})}_{\text{inv. doc. freq.}} = \log\left(\frac{|\mathcal{C}|}{\mathrm{df}(v,\mathcal{C})}\right)$$

$$\mathrm{tfidf}(v,d,\mathcal{C}) = \mathrm{tf}(v,d) \times \mathrm{idf}(v,\mathcal{C})$$

Intuition of IDF:

- Word in *every* document $\Rightarrow \mathrm{df} = |\mathcal{C}| \Rightarrow \mathrm{idf} = \log(1) = 0$ (suppressed)
- Word in *one* document $\Rightarrow \mathrm{idf} = \log|\mathcal{C}|$ (maximally discriminative)

Document similarity: cosine on ℓ_2-normalised TF-IDF vectors

$$\mathrm{sim}(d_i,d_j) = \frac{\tilde{\mathbf{x}}(d_i)^\top \tilde{\mathbf{x}}(d_j)}{\|\tilde{\mathbf{x}}(d_i)\|_2 \, \|\tilde{\mathbf{x}}(d_j)\|_2} \in [0,1]$$

Limitations of Classical Methods

Despite TF-IDF's improvements over raw counts, all classical methods share:

1. **Order invariance**
   "Earnings beat estimates" = "Estimates beat earnings" (identical BoW/TF-IDF vectors)
2. **No semantic similarity**
   Synonyms that never co-occur (*revenue*, *sales*) have zero cosine similarity
3. **Fixed vocabulary**
   New words at test time (acronyms, company names) are out-of-vocabulary and discarded
4. **High-dimensional sparsity**
   $V \sim 10^4$–10^5; even with IDF weighting, unsuitable as direct inputs to dense neural layers without dimensionality reduction

Solution: replace the discrete one-hot basis with a continuous embedding space.

The Distributional Hypothesis

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"You shall know a word by the company it keeps." — J.R. Firth (1957)

Words with similar meanings appear in similar contexts. Capture contextual co-occurrence $\Rightarrow$ capture semantic similarity.

Definition: Word Embedding

An injective map $\varphi : \mathcal{V} \to \mathbb{R}^d$ with $d \ll V$.
Embedding matrix $\mathbf{W} \in \mathbb{R}^{V \times d}$; row $i = \varphi(w_i)$.
Typical: $d \in \{50, 100, 300\}$; compression ratio $V/d \approx$ 150–1,000.

Geometry: under a well-trained embedding, cosine similarity $\approx$ semantic similarity.
Linear analogies: $\varphi(\text{king}) - \varphi(\text{man}) + \varphi(\text{woman}) \approx \varphi(\text{queen})$
In finance: $\varphi(\text{call option}) - \varphi(\text{upside}) + \varphi(\text{downside}) \approx \varphi(\text{put option})$

Word2Vec: Skip-Gram Objective

Each centre word w_t predicts context words in window $\pm c$:

$$J_{\text{SG}} = \frac{1}{T} \sum_{t=1}^{T} \sum_{\substack{k=-c \\ k\neq 0}}^{c} \log P(w_{t+k} \mid w_t)$$

where the softmax probability is:

$$P(w_{t+k} \mid w_t) = \frac{\exp\left(\mathbf{v}_{w_{t+k}}^\top \hat{\mathbf{v}}_{w_t}\right)}{\sum_{w=1}^{V} \exp\left(\mathbf{v}_w^\top \hat{\mathbf{v}}_{w_t}\right)}$$

UNDER THE HOOD

The computational problem. Normalising constant sums over all $V \approx 10^4$–10^5 words *at every gradient step*. For 1 billion tokens: intractable.

Two embedding matrices: *centre-word* vectors $\hat{\mathbf{v}}_w$ and *context-word* vectors $\mathbf{v}_w$.

Negative Sampling: Efficient Training

Replace the full softmax with a binary classification task:

- 1 *positive* sample: the observed context word w_{t+k}
- K *negative* samples: words drawn from noise distribution P_n

$$\mathcal{L}_{\text{NEG}}(w_t, w_{t+k}) = \log\sigma\left(\mathbf{v}_{w_{t+k}}^{\top}\hat{\mathbf{v}}_{w_t}\right) + \sum_{j=1}^{K}\mathbb{E}_{w_j\sim P_n}\left[\log\sigma\left(-\mathbf{v}_{w_j}^{\top}\hat{\mathbf{v}}_{w_t}\right)\right]$$

Gradient interpretation:

$$\frac{\partial\mathcal{L}_{\text{NEG}}}{\partial\mathbf{v}_{w_{t+k}}} = \left[1-\sigma\left(\mathbf{v}_{w_{t+k}}^{\top}\hat{\mathbf{v}}_{w_t}\right)\right]\hat{\mathbf{v}}_{w_t}$$

Positive context vector is pushed *toward* centre vector (large coefficient when dissimilar); negative sample vectors are pushed *away*.

Efficiency: only $K+1$ rows of $\mathbf{W}$ touched per step, not all V.
Recommended: $K = 5$–20; $P_n(w) \propto \text{freq}(w)^{3/4}$.

Finance-Specific Domain Adaptation

General embeddings (Wikipedia/Common Crawl) encode general-English meaning:

Word	General nearest neighbours	Finance nearest neighbours
risk	danger, threat, hazard	volatility, exposure, VaR
short	brief, concise	short-sell, bear position
margin	edge, border	profit margin, maintenance margin

Domain adaptation strategies:

1. **Pre-train from scratch** on financial corpus (Bloomberg Financial Word2Vec; FinancialPhraseBank embeddings)
2. **Fine-tune transformer embeddings** (FinBERT: BERT re-trained on Reuters news, 10-K filings, earnings calls)
3. **Retrofitting**: post-hoc adjustment using a financial lexicon or ontology to pull synonyms together and push antonyms apart

What We Have Covered and What Comes Next

This lecture:

- Why text predicts returns: three EMH-consistent mechanisms
- Formal textual signal framework; linear factor model; identification challenges
- Classical representations: BoW, TF-IDF, one-hot — order-invariant, discrete metric
- Word embeddings: distributional hypothesis, Word2Vec (NEG), GloVe, domain adaptation

Practical Session 1: TF-IDF by hand · cosine similarity · Python: TF-IDF on earnings call snippets **Lecture 2 (LLM Foundations):**

- RNNs: sequential order, vanishing gradients
- LSTM: cell state, forget/input/output gates
- Transformer: scaled dot-product attention, multi-head, positional encoding
- Pre-training objectives: masked LM (BERT) vs. autoregressive (GPT)

Key refs: Tetlock (2007) · LM (2011) · Mikolov (2013) · Pennington (2014) · Lopez-Lira & Tang (2023)

Appendix

Extended Examples, Timeline, and GloVe

Extra / optional material — beyond the core lecture.

From Dictionaries to Neural Methods: A Timeline

Year	Method	Key advance
1966	General Inquirer	Systematic word counting
2003	LDA (topic models)	Data-driven topics, no labels needed
2011	LM Dictionary	Financial-domain sentiment
2013	Word2Vec / GloVe	Dense embeddings, semantic geometry
1997/2015	LSTM / Attention	Sequential context, negation
2017	Transformer	Parallel self-attention, massive scale
2018–19	BERT / FinBERT	Pre-train then fine-tune
2020+	GPT-4, Claude, Llama	Instruction-following, reasoning

Each generation addresses the core limitation of the previous one.
Understanding this history explains *why* LLMs behave the way they do.

TF-IDF: Worked Example

Three pre-processed financial sentences ($N = 3$, $V = 8$):

d	Tokens
d_1	`revenue grew strong earnings`
d_2	`earnings declined weak guidance`
d_3	`revenue guidance raised strong`

$\text{idf}(\texttt{declined}) = \text{idf}(\texttt{grew}) = \text{idf}(\texttt{raised}) = \text{idf}(\texttt{weak}) = \ln(3) \approx 1.099$
$\text{idf}(\texttt{earnings}) = \text{idf}(\texttt{guidance}) = \text{idf}(\texttt{revenue}) = \text{idf}(\texttt{strong}) = \ln(3/2) \approx 0.405$

Cosine similarities

$\text{sim}(d_1, d_3) \approx 0.213$ (shared: `revenue, strong`)
$\text{sim}(d_1, d_2) \approx 0.106$ (shared: `earnings` only)

Full computation in Practical Session 1.

GloVe: Global Co-occurrence Factorisation

Co-occurrence matrix X_{ij} = times word j appears in context of word i across the full corpus. **Key insight:** the *ratio* P_{ik}/P_{jk} encodes relational meaning more faithfully than raw probabilities. GloVe minimises:

$$J_{\text{GloVe}} = \sum_{i,j=1}^{V} f(X_{ij}) \left(\mathbf{w}_i^\top \tilde{\mathbf{w}}_j + b_i + \tilde{b}_j - \log X_{ij} \right)^2$$

with weighting function $f(x) = (x/x_{\max})^\alpha$ for $x < x_{\max}$, else 1. Recommended: $\alpha = 0.75$, $x_{\max} = 100$.

GloVe vs. Word2Vec

GloVe	single-pass WLS over precomputed counts; faster with stored matrix
Word2Vec	online SGD; scales to streaming corpora

Similar embedding quality in practice.

Case Study: PCA on Financial Vocabulary

Project 30 financial terms into 2D via PCA on GloVe vectors. Colour by semantic group:
rates & macro · equities & returns · risk · corporate actions & derivatives

Consistent patterns across pre-trained models

- *option, futures, derivative* → tight cluster
- *default, credit* → cluster together, away from equities
- *merger, acquisition* → nearly identical vectors
- *alpha, beta, factor* → cluster (shared asset-pricing origin)
- Rates cluster (*yield, spread, inflation, interest rate*) → most cohesive

Caution on axis interpretation

PCA axes $\mathbf{p}_1, \mathbf{p}_2$ are *not* interpretable as named financial concepts without further validation.

Full code: `code/notebooks/01-intro/demo.ipynb`